CAMBRIDGE
International Education

Cambridge O Level

PHYSICS **5054/22**

Paper 2 Theory **May/June 2025**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct / valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standard isation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	information missing or insufficient for credit
	used to highlight parts of an extended response
	evaluation attempted
	benefit of the doubt given
	contradiction in response, mark not awarded
	incorrect point or mark not awarded
	error carried forward applied
	incorrect or insufficient point ignored while marking the rest of the response
	two statements are linked
	mandatory mark not awarded
	benefit of doubt was considered, but the response was decided to not be sufficiently close for benefit of doubt to be applied.

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Annotation	Meaning
POT	power of ten error
SEEN	point has been noted, but no credit has been given or blank page seen
SF	error in number of significant figures
TV	response is too vague or there is insufficient detail in response
	correct point or mark awarded
U	Incorrect unit

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Question	Answer	Marks
1(a)(i)	points plotted to within ½ a small square	**B1**
	smooth curve drawn	**B1**
1(a)(ii)	(average speed =) d / t in any form	**C1**
	195 (cm / s)	**A1**
1(a)(iii)	draw a tangent (at t = 0.40 s)	**B1**
	find the gradient / slope (of the graph) **or** the gradient of the tangent	**B1**
1(b)(i)	mention of air resistance / resistive force	**B1**
	resultant force decreases **or** air resistance / upwards force increases (as speed increases) / becomes high(er)	**B1**
	(eventually) air resistance = weight (and acceleration is zero) **or** air resistance balances / cancels weight	**B1**
1(b)(ii)	straight line (of positive) **or** (line of) constant gradient / slope	**B1**

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Question	Answer	Marks
2(a)	work done or energy per unit time	**B1**
2(b)	(input power =) power output / efficiency in any form	**C1**
	860 (W)	**A1**
2(c)(i)	(work =) force × distance **or** *mgh*	**C1**
	1100 (J)	**A1**
2(c)(ii)	(time =) work ÷ (output) power	**C1**
	1.9 or 1.8 s	**A1**
2(d)	*Any one of*: • friction / air resistance • electrical heating	**B1**

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Question	Answer	Marks
3(a)(i)	(P =) force / area in any form	**C1**
	1.9×10^5 (Pa)	**A1**
3(a)(ii)	(increasing pressure) decreases the area	**B1**
3(b)(i)	mass (of the particle) × velocity (of the particle)	**B1**
3(b)(ii)	*on collision*: change of momentum / direction / velocity **or** *on collision:* momentum transferred to wall	**B1**
	force related to change in momentum ÷ time **or** force of equal size on tyre / wall and on particle (Newton's 3rd law)	**B1**
3(c)	(particles have) high(er) speed / velocity / momentum / kinetic energy	**B1**
	more collisions per second **or** hit (wall) hard(er) **or** larger change in momentum per sec	**B1**

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Question	Answer	Marks
4(a)	*motion of particles:* backwards and forwards / to and fro / closer and further apart	**B1**
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork **or** mention of compressions and rarefactions	**B1**
4(b)(i)	number of oscillations per second **or** number of wave(length)s (passing a point) per second	**B1**
4(b)(ii)	2.5 (waves) seen **or** (1 wave takes) 0.02 (s) **or** (f =) 1 / T or (number of) waves / time	**C1**
	50 (Hz)	**A1**
4(b)(iii)	quieter / less volume	**B1**
4(b)(iv)	half frequency shown **and** at least one complete oscillation seen	**B1**
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	**B1**
	same wavelength as on left **and** some curving in correct direction	**B1**
4(c)(ii)	diffraction	**B1**

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Question	Answer	Marks
5(a)	swap voltmeter for ammeter	**B1**
	have cells all facing the same way, e.g. reverse left two cells	**B1**
5(b)(i)	(R =) V / I in any form, e.g. 6 / 0.8	**C1**
	7.5 (Ω)	**A1**
5(b)(ii)	shorter **and** *one of*: • resistance smaller **or** resistance proportional to length • larger current (for same voltage) • smaller voltage to produce (an equal) current	**B1**
5(b)(iii)	6 (.0 cm)	**B1**

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Question	Answer	Marks
6(a)	(I) = P / V **or** 2000 / 230 **or** 8.7 (A)	**M1**
	fuse rating 13 A underlined or circled	**A1**
	The fuse rating needs to be the lowest value above / slightly larger than the normal current / 8.7 A **or** 5 / 8 A fuse blows under normal operation / 8.7 A **or** 30 A fuse may not blow / heater or circuit damaged if fault or current larger than normal 8.7 A	**B1**
6(b)	fuse melts / blows / **or** cuts off current / circuit	**B1**
6(c)(i)	0.95 (A)	**B1**
6(c)(ii)	17	**B1**
6(d)	*Any two from*: (trip switches can / are) • be reset / used more than once • faster acting / turns off immeditaely • more sensitive • be tested without replacement • have an external indication that they are working • can trip on difference in live and neutral currents	**B2**

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Question	Answer	Marks
7(a)	*split ring commutator*: to reverse the current (in the coil)	**B1**
	reverse current / connections every half turn / twice in one turn / every 180° / when coil is vertical	**B1**
	brushes: to connect coil / commutator or (split) rings to battery / (external) circuit **or** to provide current to coil, commutator or (split) rings **or** to avoid wires (to coil) tangling / twisting	**B1**
7(b)	*direction*: (from left) to right / from magnet 1 to magnet 2	**B1**
	explanation: current is from plus to minus / towards front / out of plane / page / outwards / top to bottom of wire / downwards along wire / towards the split ring on the left	**B1**
	left-hand rule mentioned and explained using two of current, field or motion / force	**B1**
7(c)	no change	**B1**
7(d)(i)	(moment =) force × distance in any form	**C1**
	0.69	**A1**
	Nm	**B1**
7(d)(ii)	(perpendicular) distance between (line of action of) forces decreases **or** (sides of) coil enter weaker magnetic field **or** coil leaves magnetic field	**B1**

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Question	Answer	Marks
8(a)	*shielding*: protects workers / absorbs radiation	**B1**
	control rods: controls number of neutrons / absorbs neutrons	**B1**
	moderator: slows down the neutrons	**B1**
8(b)	neutron hits / collides U–235 (nucleus)	**B1**
	nucleus splits / forms daughter nuclei	**B1**
	emits more neutrons	**B1**
8(c)(i)	amount of energy / heat needed for a unit rise in temperature	**M1**
	for unit mass	**A1**
8(c)(ii)	large (specific heat capacity) **and** *one of*: • smaller rise in temperature / temp. stays lower • heats up slower / longer time • less chance of boiling • smaller mass needed (for same temperature rise) / less flow rate / smaller pumps • more energy absorbed <u>for same temp. rise</u> **or** small(er heat capacity) **and** larger temperature rise (to heat boiler more efficiently) / heats up faster	**B1**

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Question	Answer	Marks
9(a)	Mercury orbit inside Earth's orbit	**B1**
	Mars orbit between orbit of Earth and Jupiter	**B1**
9(b)	the Sun is in the way	**B1**
9(c)(i)	gravitational (force / attraction from Sun)	**B1**
9(c)(ii)	all four forces towards Sun by eye	**B1**
9(c)(iii)	increases (from 1) to 3 **and** decreases (from 3 to 4)	**B1**
	explanation of statement in description e.g. (stronger force) because comet is close(r) to Sun (at position 3)	**B1**
9(d)	speed = $2\pi r / T$ in any form	**C1**
	time for Jupiter to orbit Sun = 3.8×10^8 (s) **or** time for Earth to orbit Sun = 3.1×10^7 (s) **or** distance increases by 7.8 / 1.5 **and** speed decreases by 30 / 13 **or** $\frac{7.8\,(\times 10^8)}{1.5\,(\times 10^8)} \times \frac{30}{13}$	**C1**
	number of orbits = 12	**A1**

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